Institute of Technology of Cambodia

Department of Information and Communication Engineering

Assignment: TP06

Subject: **Operating System**

Lecturer Course/TP: **Heng Rathpisey**

Student: **Saren Sokmeak**

ID: **e20211376**

Year: **I4**

Academic year: 2024-2025

Task 1: Running and Monitoring Processes

1. Setup:

a. Write three Python scripts: cpu_intensive.py, memory_intensive.py, and long_running.py.

```
sokmeak@sokmeaksaren:~/TP6$ ls
cpu_intensive.py  long_running.py  memory_intensive.py
sokmeak@sokmeaksaren:~/TP6$
```

```
import time
def cpu_intensive_task():
        while True:
                sum([i**2 for i in range(10000)])

if __name__ == "__main__":
 while True:
  cpu_intensive_task()
  time.sleep(1)
```

```
import time
def memory_intensive_task():
        large_list = [0] * 10**7
        time.sleep(30)

if __name__ == "__main__":
 while True:
  memory_intensive_task()
```

```
import time
if __name__ == "__main__":
 while True:
  print("Running...")
  time.sleep(5)
```

b. Run each script: python3 [script_name].py &.

```
sokmeak@sokmeaksaren:~/TP6$ python3 cpu_intensive.py
```

```
sokmeak@sokmeaksaren:~/TP6$ python3 memory_intensive.py
```

```
sokmeak@sokmeaksaren:~/TP6$ python3 long_running.py
Running...
Running...
```

2. Monitoring:

a. Use ps aux to list the processes. and

```
sokmeak@sokmeaksaren:~/TP6$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.3  23212 13092 ?        Ss   10:30   0:03 /sbin/init
root           2  0.0  0.0      0     0 ?        S    10:30   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    10:30   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   10:30   0:00 [kworker/R-rcu_g]
root           5  0.0  0.0      0     0 ?        I<   10:30   0:00 [kworker/R-rcu_p]
root           6  0.0  0.0      0     0 ?        I<   10:30   0:00 [kworker/R-slub_]
root           7  0.0  0.0      0     0 ?        I<   10:30   0:00 [kworker/R-netns]
root           8  0.0  0.0      0     0 ?        I    10:30   0:02 [kworker/0:0-events]
root           9  0.0  0.0      0     0 ?        I<   10:30   0:00 [kworker/0:0H-events_highpri]
root          12  0.0  0.0      0     0 ?        I<   10:30   0:00 [kworker/R-mm_pe]
root          13  0.0  0.0      0     0 ?        I    10:30   0:00 [rcu_tasks_kthread]
root          14  0.0  0.0      0     0 ?        I    10:30   0:00 [rcu_tasks_rude_kthread]
root          15  0.0  0.0      0     0 ?        I    10:30   0:00 [rcu_tasks_trace_kthread]
root          16  0.0  0.0      0     0 ?        S    10:30   0:00 [ksoftirqd/0]
root          17  0.0  0.0      0     0 ?        I    10:30   0:00 [rcu_preempt]
root          18  0.0  0.0      0     0 ?        S    10:30   0:00 [migration/0]
root          19  0.0  0.0      0     0 ?        S    10:30   0:00 [idle_inject/0]
root          20  0.0  0.0      0     0 ?        S    10:30   0:00 [cpuhp/0]
root          21  0.0  0.0      0     0 ?        S    10:30   0:00 [cpuhp/1]
root          22  0.0  0.0      0     0 ?        S    10:30   0:00 [idle_inject/1]
root          23  0.0  0.0      0     0 ?        S    10:30   0:00 [migration/1]
root          24  0.0  0.0      0     0 ?        S    10:30   0:00 [ksoftirqd/1]
root          26  0.0  0.0      0     0 ?        I<   10:30   0:00 [kworker/1:0H-events_highpri]
root          27  0.0  0.0      0     0 ?        S    10:30   0:00 [cpuhp/2]
root          28  0.0  0.0      0     0 ?        S    10:30   0:00 [idle_inject/2]
root          29  0.0  0.0      0     0 ?        S    10:30   0:00 [migration/2]
root          30  0.0  0.0      0     0 ?        S    10:30   0:00 [ksoftirqd/2]
root          32  0.0  0.0      0     0 ?        I<   10:30   0:00 [kworker/2:0H-events_highpri]
root          33  0.0  0.0      0     0 ?        S    10:30   0:00 [cpuhp/3]
root          34  0.0  0.0      0     0 ?        S    10:30   0:00 [idle_inject/3]
root          35  0.0  0.0      0     0 ?        S    10:30   0:00 [migration/3]
root          36  0.0  0.0      0     0 ?        S    10:30   0:00 [ksoftirqd/3]
root          38  0.0  0.0      0     0 ?        I<   10:30   0:00 [kworker/3:0H-kblockd]
root          39  0.0  0.0      0     0 ?        S    10:30   0:00 [cpuhp/4]
```

b. Observe CPU and memory usage with top or htop.

```
  0[                                                          0.0%]   3[                                                          0.0%]
  1[||                                                        1.9%]   4[||||||||||||||||||||||||||||||||||||||||||||||||||||||||||94.0%]
  2[||||                                                      6.0%]   5[                                                          0.0%]
Mem[||||||||||||||||||||||||||||||                       931M/3.81G] Tasks: 123, 369 thr, 111 kthr; 2 running
Swp[                                                      0K/3.81G] Load average: 0.84 0.39 0.33
                                                                    Uptime: 00:54:38

  Main   I/O
    PID USER       PRI  NI  VIRT   RES   SHR S  CPU%▽MEM%   TIME+  Command
   2871 sokmeak     20   0  485M 27652 20868 S   0.0  0.7  0:00.09 /usr/bin/update-notifier
   2883 sokmeak     20   0  485M 27652 20868 S   0.0  0.7  0:00.00 /usr/bin/update-notifier
   2884 sokmeak     20   0  485M 27652 20868 S   0.0  0.7  0:00.21 /usr/bin/update-notifier
   2886 sokmeak     20   0  485M 27652 20868 S   0.0  0.7  0:00.00 /usr/bin/update-notifier
   2887 sokmeak     20   0  485M 27652 20868 S   0.0  0.7  0:00.00 /usr/bin/update-notifier
   3009 sokmeak     20   0  701M 66576 48808 S   0.0  1.7  0:00.02 /usr/libexec/gnome-terminal-server
   3028 sokmeak     20   0  8504  4992  3456 S   0.0  0.1  0:00.11 bash
   3050 sokmeak     20   0 17352  9728  5760 T   0.0  0.2  0:45.65 python3 cpu_intensive.py
   3070 sokmeak     20   0 17252  9344  5760 S   0.0  0.2  0:00.11 python3 long_running.py
   3196 sokmeak     20   0 2946M 61668 47256 S   0.0  1.5  0:00.33 gjs /usr/share/gnome-shell/extensions/ding@rastersoft.com/app/ding.js -E -P
   3200 sokmeak     20   0 2946M 61668 47256 S   0.0  1.5  0:00.00 gjs /usr/share/gnome-shell/extensions/ding@rastersoft.com/app/ding.js -E -P
   3201 sokmeak     20   0 2946M 61668 47256 S   0.0  1.5  0:00.00 gjs /usr/share/gnome-shell/extensions/ding@rastersoft.com/app/ding.js -E -P
   3202 sokmeak     20   0 2946M 61668 47256 S   0.0  1.5  0:00.02 gjs /usr/share/gnome-shell/extensions/ding@rastersoft.com/app/ding.js -E -P
   3203 sokmeak     20   0 2946M 61668 47256 S   0.0  1.5  0:00.01 gjs /usr/share/gnome-shell/extensions/ding@rastersoft.com/app/ding.js -E -P
   3204 sokmeak     20   0 2946M 61668 47256 S   0.0  1.5  0:00.00 gjs /usr/share/gnome-shell/extensions/ding@rastersoft.com/app/ding.js -E -P
   3205 sokmeak     20   0 2946M 61668 47256 S   0.0  1.5  0:00.00 gjs /usr/share/gnome-shell/extensions/ding@rastersoft.com/app/ding.js -E -P
   3206 sokmeak     20   0 2946M 61668 47256 S   0.0  1.5  0:00.00 gjs /usr/share/gnome-shell/extensions/ding@rastersoft.com/app/ding.js -E -P
   3207 sokmeak     20   0 2946M 61668 47256 S   0.0  1.5  0:00.00 gjs /usr/share/gnome-shell/extensions/ding@rastersoft.com/app/ding.js -E -P
   3208 sokmeak     20   0 2946M 61668 47256 S   0.0  1.5  0:00.00 gjs /usr/share/gnome-shell/extensions/ding@rastersoft.com/app/ding.js -E -P
   3218 sokmeak     20   0 2946M 61668 47256 S   0.0  1.5  0:00.00 gjs /usr/share/gnome-shell/extensions/ding@rastersoft.com/app/ding.js -E -P
   3243 sokmeak     20   0 2946M 61668 47256 S   0.0  1.5  0:00.00 gjs /usr/share/gnome-shell/extensions/ding@rastersoft.com/app/ding.js -E -P
   3325 sokmeak     20   0 17252  9344  5760 S   0.0  0.2  0:00.03 python3 long_running.py
   3357 sokmeak     20   0  8504  4992  3456 S   0.0  0.1  0:00.01 bash
   3371 sokmeak     20   0  8504  5120  3456 S   0.0  0.1  0:00.01 bash
   3384 sokmeak     20   0 95380 87524  5760 S   0.0  2.2  0:00.06 python3 memory_intensive.py
   3385 sokmeak     20   0  8372  4992  3456 S   0.0  0.1  0:00.01 bash
F1Help  F2Setup F3SearchF4FilterF5Tree  F6SortByF7Nice -F8Nice +F9Kill  F10Quit
```

Task 2: Managing Process Priority

3. Prioritization:

a. Identify the PID of cpu_intensive.py.

```
   3050 sokmeak     20   0 17352  9728  5760 T   0.0  0.2  0:45.65 python3 cpu_intensive.py
```

b. Lower its priority (increase niceness) using renice.

Priority range: -20 (highest prio) : 19 (lowest prio)

- nice -n 10 python3 cpu_intensive_1.py &

```
sokmeak@sokmeaksaren:~/TP6$ nice -n 10 python3 cpu_intensive_1.py &
[1] 3522
sokmeak@sokmeaksaren:~/TP6$
```

- nice -n -1 python3 cpu_intensive_2.py &

```
sokmeak@sokmeaksaren:~/TP6$ nice -n -1 python3 cpu_intensive_2.py &
[2] 3524
[1]   Done                    nice -n 10 python3 cpu_intensive_1.py
sokmeak@sokmeaksaren:~/TP6$ nice: cannot set niceness: Permission denied
```

- renice -n -10 -p PID of cpu_intensive_1.py &

```
sokmeak@sokmeaksaren:~/TP6$ sudo renice -n -10 -p 3050
[sudo] password for sokmeak:
Sorry, try again.
[sudo] password for sokmeak:
3050 (process ID) old priority 0, new priority -10
sokmeak@sokmeaksaren:~/TP6$
```

- renice -n 5 -p PID cpu_intensive_2.py &

```
sokmeak@sokmeaksaren:~/TP6$ renice -n 5 -p 3493 cpu_intensive_2.py &
[1] 3554
sokmeak@sokmeaksaren:~/TP6$ 3493 (process ID) old priority 0, new priority 5
renice: bad process ID value: cpu_intensive_2.py
```

Task 3: Process Termination

4. Termination:

a. Use kill to terminate long_running.py with SIGTERM and SIGKILL.

```
  3050 sokmeak     10 -10 17352  9728  5760 T   0.0  0.2  0:45.65 python3 cpu_intensive.py
  3070 sokmeak     20   0 17252  9344  5760 S   0.0  0.2  0:00.14 python3 long_running.py
  3196 sokmeak     20   0 2946M 61668 47256 S   0.0  1.5  0:00.34 gjs /usr/share/gnome-shell/extensions/ding@rastersoft.com/app/ding.js
  3200 sokmeak     20   0 2946M 61668 47256 S   0.0  1.5  0:00.00 gjs /usr/share/gnome-shell/extensions/ding@rastersoft.com/app/ding.js
```

```
sokmeak@sokmeaksaren:~/TP6$ kill 3070
```

```
  3494 sokmeak     20   0  8372  5120  3584 S   0.0  0.1  0:00.01 bash
  3510 sokmeak     20   0  8632  5248  3456 S   0.0  0.1  0:00.03 bash
  3555 sokmeak     20   0  8372  4992  3456 S   0.0  0.1  0:00.01 bash
  3580 sokmeak     20   0 17252  9344  5760 S   0.0  0.2  0:00.01 python3 long_running.py
```

```
sokmeak@sokmeaksaren:~/TP6$ kill -9 3580
```

Task 4: Resource Management 5. Resource Observation:

a. Determine which script is most resource-intensive using top or htop.

```
Main   I/O
  PID USER      PRI  NI  VIRT   RES   SHR S  CPU%▽MEM%   TIME+  Command
 3597 sokmeak    20   0 17352  9728  5760 R 100.3  0.2  6:52.44 python3 cpu_intensive.py
```

+ **cpu_intensive.py** is the most resource-intensive by using htop.